

Unleashing the intranet

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Intranets are an increasingly important part of corporate information networks — but in a move to the full service intranet, where the whole business can be operated from its confines, there are many obstacles. As organisations now realise the basics of an intranet service, BT is tasked with providing a ‘total’ solution to both new innovative products and legacy applications alike. This paper describes the technologies underlying intranets and looks at the possible effects of emerging technologies such as network computing and the supporting infrastructure for collaboration.

1. Introduction

Many people have heard of or used the Internet. An intranet is nothing more complicated than an internal network which makes use of Internet technologies based on TCP/IP (transmission control protocol/ internet protocol). Originally attributed to the Wall Street Journal back in November 1995, who coined the term, intranets have now become a major sales sector in their own right. Zona Research of Redwood, California assessed the intranet market to be worth \$476 million in 1995 and predicted it will reach \$4 billion by 1997. This year they revised their predictions and announced sales of \$6 billion for intranets, \$500 million more than for the Internet [1]. Essentially all that is required for a simple intranet is a network running the TCP/IP stack with a webserver generating the information content and a Web browser client. In order to separate this from the Internet, security is needed in firewalls [2], creating an enclosed area which allows users to interact with each other and to access the Internet if allowed, while keeping potential intruders out.

For sectors such as the utilities or defence, the intranet may not even be connected to the Internet. Organizations who are connected may also limit the number of users who can access the Internet. The user interface is provided by standard Web browsers supporting HTTP such as Netscape Navigator and Microsoft Internet Explorer.

Intranets (see Fig 1) offer many benefits over existing networks derived from the universal communication made possible by the use of TCP/IP and other Internet protocols. Information can be shared irrespective of the hardware platform being used. Using this ability and the relatively high bandwidth that an internal network offers, new applications are possible such as videoconferencing, multimedia and audio communications. Intranets provide for:

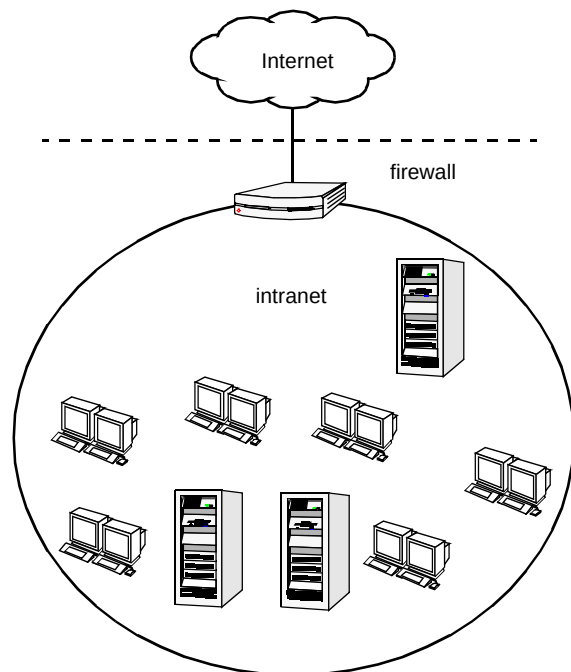


Fig 1 The intranet concept.

- ease of communication,
- cross-platform functionality,
- consistent view of information,
- ease of integration,
- enabling technology (e.g. for videoconferencing and multimedia),
- collaboration with partners and suppliers,
- increased effectiveness of information,
- relatively low cost of installation.

This is only possible through the cross-party support for open standards which is one of the reasons why Internet technologies are advancing so quickly. Standards such as HTTP (hypertext transmission protocol) and HTML (hypertext mark-up language) ensure that information need only be published once and can be read on almost any hardware platform currently available through browsers like Microsoft Internet Explorer and Netscape Navigator.

Another potential area of growth is that of the extended intranet or extranet. This is an enclosed space between intranets, which allows companies to collaborate on joint projects. The extranet offers the benefits of electronic interaction between employees in a virtual private network enclosed from the Internet. Conceptually the basic form of an extranet is shown in Fig 2. A shared space is provided by a file server which can be accessed by two or more parties from outside their own intranet. A more logical idea would be to configure server spaces within the intranet to which the other company has access, but from which they cannot access other sensitive information. Using traditional client/server computing technologies this would lead to problems over intellectual property rights and disputes over control of the information. Despite market predictions that extranets will form up to 20% of corporate effort by the next millennium [1] these obstacles hamper their implementation. Firewalls also tend to provide an 'all-or-nothing' defence which makes the management of restricted access very difficult. This offers opportunities for telecommunications companies to provide hosting services on 'neutral' ground. However, there are also moves to change the way business

is conducted over the Internet which more directly may allow such co-operation.

2. Intranet security

Security across the Internet is of the utmost concern to all businesses and is one of the reasons that the huge profit that companies anticipate from trading over the Internet is currently confined mainly to advertising. Within the intranet environment security is provided by the same mechanisms as the underlying client/server network, upon which most are built. This uses firewalls, software programs which reside on servers or routers, and allows transactions normally based on the IP address. External Internet-facing World Wide Web (WWW) servers are usually kept in isolation or with highly secure links to the main network. Within an organisation, departmental documentation can also be secured by using features provided by popular web-servers such as the Enterprise server from Netscape and IIS from Microsoft.

Allowing business partners or individuals from different departments to gain access to an extranet, offers considerable benefits. The use of the public Internet to create virtual private networks (VPNs) is one method which would drive down costs against leased lines. This relies not only on users being certified, but also more crucially on Web sites as well, so that a mutual trust is shared between companies, digital certificates contain information about the user and also about the author of software downloaded. The emphasis on trust either directly or via a neutral authentication agency such as Verisign in the US [3] may be adequate for simple, small-scale transactions, but it poses

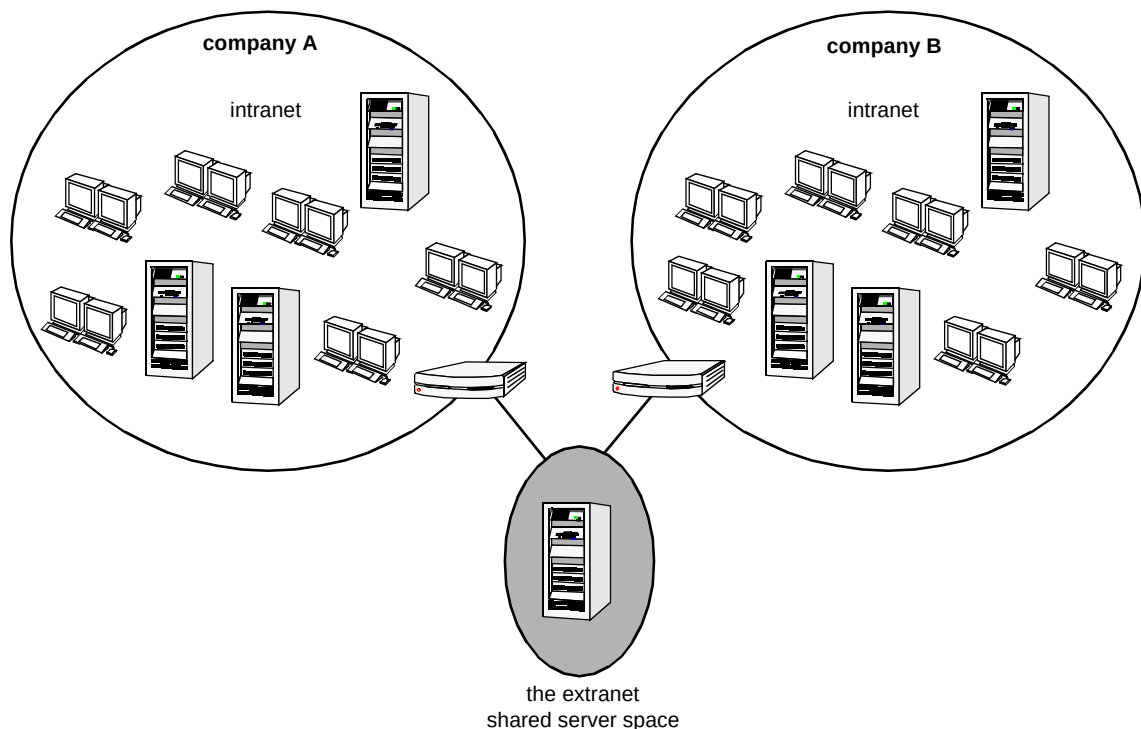


Fig 2 The extranet shared server space.

possible problems for commerce over the extranet as it could allow the use of malicious software to control the host system, providing the author is registered as a trusted party. Quality of service (QoS) within a VPN relies on an Internet service provider (ISP) such as the Concert Internet Plus product. For applications which require high levels of reliability or security, public Internet use, as a vehicle for VPNs, is inappropriate. This is even more apparent when using multicast IP.

The secure sockets layer (SSL) from Netscape (supported by Microsoft, FTP and NetManage among others) uses a two-layer system on top of TCP and allows the server and client to authenticate each other. SSL provides message privacy by encryption, integrity by authentication codes and mutual authentication by X.509 certificates. For second generation intranets, users would have digital certificates allowing access to various servers within the network. Different security profiles can also be provided, so that for example a basic level would allow collaboration between users of different departments. Certificate and directory servers could also be built to represent a corporate certification authority, serving digital certificates to business partners as well as new and existing employees. Certificates are then used via SSL-compliant Web browsers, such as Netscape Navigator and Internet Explorer (versions 3.0 and above). Advantages to the user include a single user log-in across multiple servers, secure e-mail, authentication and transaction integrity while the network administrator can monitor users' activities enterprise-wide. MasterCard and Visa are basing their Secure Transaction Standard (SET) for commercial banking over the Internet on X.509 in conjunction with IBM, Microsoft, Netscape and GTE. Microsoft have also developed their own standard called the point-to-point tunnelling protocol (PPTP) [4]. Based on PPP (point-to-point protocol), part of the TCP/IP family, PPTP is implemented in version 4.0 of Microsoft NT networking server software and can provide host-to-network and router-to-router connections. US government restrictions which limit the export of encryption keys over 56 bits apply, however, there has been some movement recently including the decision to allow US firm Open Market to export its electronic commerce software with 128-bit encryption [5].

An area of great concern has been the ability of Web browsers to access and download applets or controls which are software held centrally and downloaded within an HTML page providing added functionality. Such software may then have free access to network drives from the client and transmit information back across the firewall to a remote host. Java, a distributed object-oriented language from Sun Microsystems, guards against this by reducing the network functionality available to Java applets. Java is also run in a protected environment, called a sandbox, so that downloaded software does not have direct access to the client processor memory stack. Byte codes of information

are checked and then executed by a security management program within the Java virtual machine.

ActiveX consists of a series of technologies based at client and server level (see Fig 3). ActiveX controls use a method of certification by an encoded digital certificate bearing the authoring information. However, it is possible to leave an ActiveX control unsigned and providing the security checking software within Internet Explorer is switched off the control can still be downloaded within a Web page. As ActiveX controls are designed to be reused within Web pages or Windows applications, they also have direct access, not only to network drives, but also to the PC itself which creates an additional potential security risk. Currently ActiveX is supported only by Microsoft which requires the client browser to be Internet Explorer 3.0 or later. Active Server provides dynamic content generation and server-side scripting within IIS running on NT. This is secure as all processing is done before the HTML is sent to the client browser, thus providing a platform-independent implementation. Active Server also has the benefit of building upon the security model of NT and, with firewalling software from the likes of Checkpoint Inc, this may be extended to other parts of the network. Through SQL server access to legacy systems using the IIS model and functionality is also possible.

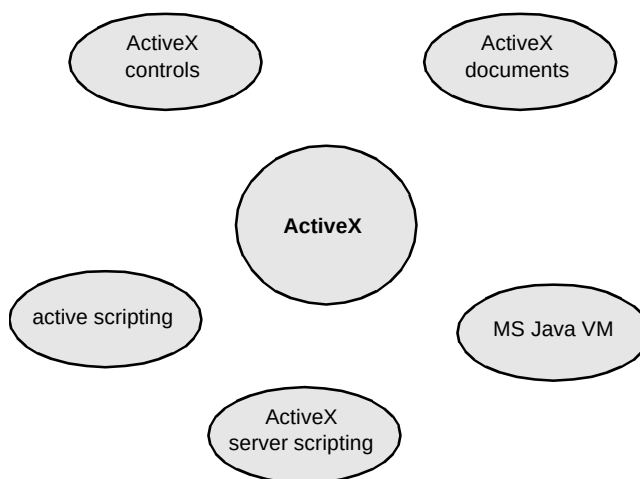


Fig 3 Microsoft's ActiveX model.

Java is supported across many platforms and by both Netscape and Internet Explorer and allows the developer to write software once and distribute across all platforms within the enterprise. JavaBeans promises to provide ActiveX like components using Java and without the same security implications. Sun originally decided that if certain classes within the Java Development Kit (JDK) could not be guaranteed to be safe for use they would be disabled until a later date. Now at version 1.1 the JDK provides a series of packages which can be used to program diverse Java applications from CTI (computer/telephony interface) to enhanced Web applets and JavaBeans, although currently both Netscape and Internet Explorer support version 1.02 of

the JDK in their VMs. JavaBeans is not as widely available as ActiveX and is created solely from Java where as ActiveX controls can be created from many different development languages such as Visual Basic 5.0, Visual C++ 4.2 and J++ 1.1. However, JavaBeans allows digital certification in addition to the applet security model. An unsigned JavaBean has access only to basic applet functionality, while a trusted JavaBean has full access to functionality provided, until now, by a stand-alone Java application [6]. ActiveX controls are also platform-specific. There are varying implementations of the Java VM across platforms, stemming from Sun's decision to allow licensees to add their own functionality on top of the JDK. Netscape also makes use of browser plug-ins within its software which means that, unlike ActiveX, software is not downloaded automatically without the user's permission.

3. Intranet applications

BT's Intranet with over 80 000 connected users is one of the largest in the world and also shows many of the organisational processes needed to build a successful intranet. Typical applications for an intranet include:

- electronic mail,
- electronic publishing,
- software distribution,
- information retrieval,
- conferencing (text, audio and video),
- file transfer,
- distributed computing (ActiveX and Java),
- telephony (Internet phone and CTI),
- electronic commerce.

Just as the Internet is disorganised, the corporate intranet should be the opposite. Intranet applications and content can be split into three categories [7]:

- level 1 — operations and management information,
- level 2 — internal communications and information, e.g. technical resources,
- level 3 — local department and users' personal pages.

The first level is critical to the business, while level two provides useful information to employees. Information content should be owned and reviewed at regular intervals and should follow a predefined corporate style. Information which is accessed by many employees needs to be easily navigable and well presented. The lower down the structure, the less need there is for 24-hour provision of access and the requirements for information revision can be less rigid. Local information (level 3) tends to be of value to a few individuals during working hours. Selling an intranet to end users relies on providing useful information and providing

reference manuals or the corporate newspaper on the intranet provides a useful learning curve for IT support staff while ensuring the support of end users.

The light directory access protocol (LDAP) is a light weight open standard interface to OSI (open systems interface) X.500 systems. LDAP is designed to allow a directory hierarchy to be built upon the organisational structure and provides an interface to legacy systems as well as allowing directories to be distributed and replicated. LDAP directories can contain employee information or more complex data such as digital certificates. Microsoft have recently announced an API which sits above LDAP providing a simplified interface to directories meeting the standard as well as NT 4.0 PDC (NT 5 alpha release), Netscape Directory Server, Exchange 5.0, MCIS (Microsoft Commercial Internet System), Novell LDAP-enabled NDS, and the University of Michigan's SLDAP server, where LDAP was invented. This will allow developers to write simplified interfaces to directories using C++, J++ (Microsoft's implementation of Java) or higher level languages such as Visual Basic [8].

Microsoft has provided compatibility with existing systems by the use of ActiveX controls, which can be re-used based on the Component Object Model (COM) which has also been added to with DCOM, a method of distributing COM objects, like Java's Remote Method Invocation (RMI). Microsoft's implementation of the Java language contains enhancements which, however, often make the code unusable on other implementations of the Java VM.

Traditional groupware technologies such as Lotus Notes are moving towards greater integration with the intranet in an enterprise environment. Notes, for example, uses Domino to provide an HTTP server creating dynamic HTML. Lotus (IBM) have also announced that they will open the, until now proprietary, Notes database to Java. Using HTTP and Internet protocols it is possible to connect to virtually any system and Java will soon open up systems like IBM mainframes, HP data centres and Notes databases to the programmer. Intranets also provide a means of providing relatively inexpensive multimedia. This can either be in the form of Shockwave applets which are based on the Macromedia Director multimedia authoring tool (Fig 4) [9] or more advanced VRML worlds and video conferencing. Full multimedia content can also be provided as a separate application from the Web browser using TCP/IP as the underlying transport mechanism.

In a move to solve some of the problems associated with interoperability, the Object Management Group (OMG) announced the birth of the object browser which incorporates the Internet Inter-ORB Protocol (IIOP) as a way of accessing distributed object-oriented applications across the Internet. Netscape announced their support through the inclusion of Visigenic's Java-based object

request broker (ORB) technology in their next generation communicator and constellation products. IIOP is part of the Common Object Request Broker Architecture (CORBA) and is supported by more than 600 companies [10]. By enabling an object-oriented common API, intranet and future extranet applications will be able to interact across corporate boundaries.

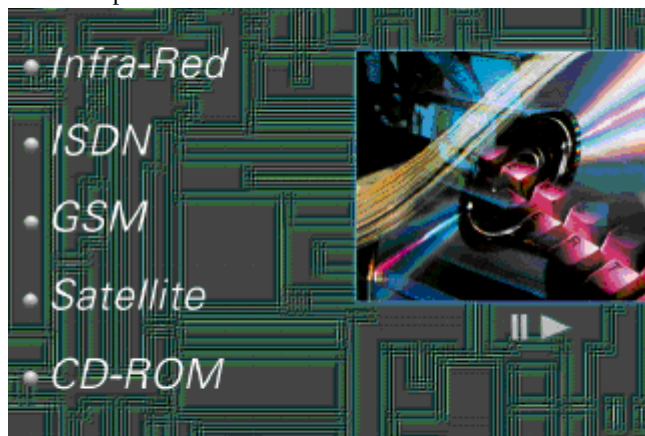


Fig 4 Multimedia presentation client written using Director 5.0 [9].

4. The push revolution — publish and subscribe

Traditionally information has been accessed from the World Wide Web via a request and reply mechanism, also known as pull. The client initiates a request for information and the server replies. This works well for 'surfing', but for rapidly changing internal information or items of interest the user only wants to be notified when the data has changed. This requires a push information model or more ideally publish and subscribe. The user subscribes to information either by narrowed down selection or by profile, and as information becomes available on the server that matches

the user's interests it is 'pushed' out to the client. Billed as a new 'killer' application, publish and subscribe technologies have the potential to improve intranet applications particularly in areas like marketing and sales, employee communications and technical briefings.

Current products on the market tend to fit into three areas, application content, middleware (application-level multicasting) and network-level multicasting. Nearly all of the applications at the content level use a sophisticated pull model which incorporates client software with advanced features rather than running through a Web browser. The client polls the remote server and requests information from content channels to which the user has subscribed. This creates the illusion of publish and subscribe, but is mainly focused at news feeds and marketing applications. Middleware from companies like TibCo Inc and Iona Technologies (Orbix), has been around for a few years within the financial services industry but relies on a directory of current users and uses a unicast transmission to push information out to the client almost like an e-mail list. Network manufacturers are working on reliable multicast IP which will allow highly efficient use of the bandwidth available as shown in Fig 5.

Research is being conducted into reliable multicasting as UDP (user datagram protocol) is connectionless and does not provide a reliable connection, so if packets are lost they are not re-transmitted. This leads to the same quality of service (QoS) arguments that are being applied to ATM (asynchronous transfer mode) networks. For mission critical intranets the network has to be robust and for certain applications needs to provide QoS. ATM provides this by using virtual connections, a hybrid data and voice approach to networking which provides high scalability over fast

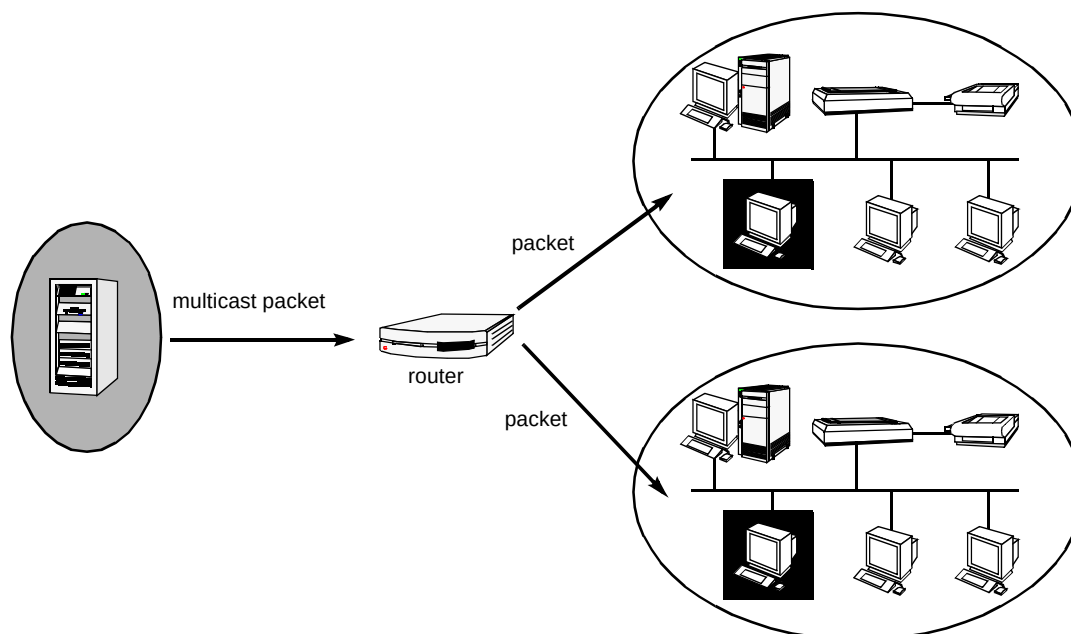


Fig 5 Multicast IP transmission.

Ethernet or FDDI (fibre distributed data interface) by not sharing throughput between users. Multicasting relies on intelligence being built into routers. A single packet is transmitted from the source and, when it reaches a router responsible for a LAN segment with a recipient client, the router copies the packet and sends it to each client. This vastly reduces the network bandwidth taken up by data transmission in comparison to the traditional request-and-reply model.

5. The impact of network computing and information profiling

Announced during November 1995, the concept of network centric computing from Oracle, Netscape, Sun and IBM has sparked a remarkable turn around from the established desktop manufacturers Intel and the software giant Microsoft. Initially portrayed as a poor man's relation to the PC with a more limited scope, Intel and Microsoft have recently announced their own NetPC profile. Californian research company Input, predict a growth in sales from \$500 million this year to \$12.5 billion in the year 2001 for the network computer (NC) [11, 12]. NCs use the network to do all the hard work, acting as a thin client. Software is loaded from a server on boot-up and reduces maintenance costs typically associated with PCs although there are anticipated increased server and network maintenance costs [11]. Market research shows that PCs will only be moderately affected, with NCs being used mainly by those who currently do not use a PC, or existing mainframe users. Input also cite the success of NCs being closely tied to the ability to migrate existing applications and systems such as Microsoft Office and database connectivity. By using a common interface relayed from the server and integration with the Java OS applications can be written for many different devices which will interact. For example, PowerTV has produced a Web TV NC, while SUN has produced the Java Station and Nokia recently announced their intention to include Java in their Communicator product, therefore making a mobile NC.

The NetPC profile is essentially a PC with all the trimmings missing [13]. Many manufacturers such as IBM and Compaq are strategically placed in both camps with a vision that data entry and clerical staff will use NCs while power users will use the NetPC and PCs. The NC is less specific on an operating system. The NetPC is based on Windows 95 and does not provide support for plug and play or for the lightweight Windows CE which was announced at the end of 1996 for the PDA market. Although there are middleware software products available which enable existing applications such as MS Word to be used on an NC, until more vendors like Corel port their software to a Java environment, corporate organisations are likely to refrain from taking the leap. This may also give the NetPC a boost as Office and other traditional applications will be supported. The NetPC will also be closely positioned with the next generation NT 5.0 server. Network centric

computing relies heavily on server scalability and power as well as the available bandwidth of the network. This will mean an even greater emphasis on a scalable infrastructure with technologies such as ATM and multicast ability.

By using security information provided by LDAP directories and digital certificates, it is possible to tailor information to suit the user's profile (see Fig 6). However, through new Internet protocols such as the IMAP4 messaging protocol, the next generation intranet should be able to provide information content based on the type of connection. BTMagic [7] provides a single contact number which can be diverted to any other number for e-mail, fax and voice. System management can also be performed not only over the Web, but also using a telephone and voice recognition.

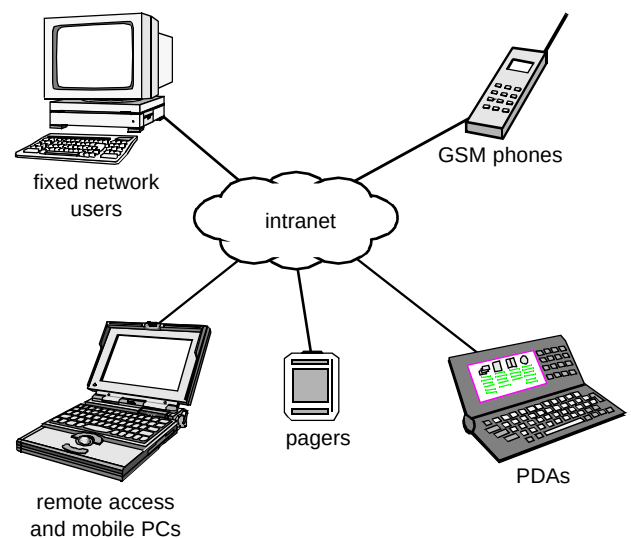


Fig 6 Intranet information profiling.

IMAP4, a messaging protocol, was developed by the University of Washington, having been first proposed by the Stanford University. It is a superset of POP (Post Office Protocol) and covers the three main types of connection — off-line, on-line and disconnected. It offers increased functionality for mobile users in particular, including the use of remote folder manipulation, multiple folder support and on-line optimisation [13]. It is also supported in the forth coming Netscape Communicator Web browser. Techniques like publish and subscribe combined with information profiling together provide the ability to give the user the right information at the right time in the right place.

6. Conclusions

Intranets have been remarkably successful in providing information of a corporate nature. In the move to the mission-critical intranet where all trading is conducted enterprise-wide, the network infrastructure is of the utmost

importance. The advent of network centric computing has also realised the potential of all devices on the network to interoperate whether they be data terminals, GSM phones, workstations or PCs. However, the need for open standards is even more crucial and object-oriented technology such as Java and CORBA-based interfaces make system integration possible right from marketing and sales, publish and subscribe to interfaces and legacy systems. In a sense the intranet has moved on from being an internal facing corporate Internet to an enterprise-wide network providing communication at all levels between distributed work groups operating in collaboration. Intranets are billed as being cheap and cost effective, and to a certain extent they are, but to operate a business or mission-critical intranet, it is of great importance that the appropriate network infrastructure is in place, particularly given the advent of network computing.

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